

AWS



ALB vs NLB

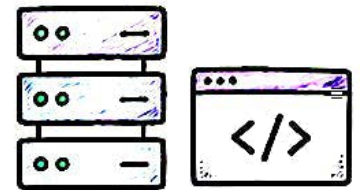


100,000 REQUESTS/SECOND
ARE HITTING YOUR APPLICATION!

WHICH LOAD BALANCER ?
WOULD YOU CHOOSE?



USERS



APPLICATION

(A)

ALB

APPLICATION
LOAD BALANCER
(LAYER 7)

(B)

NLB

NETWORK
LOAD BALANCER
(LAYER 4)



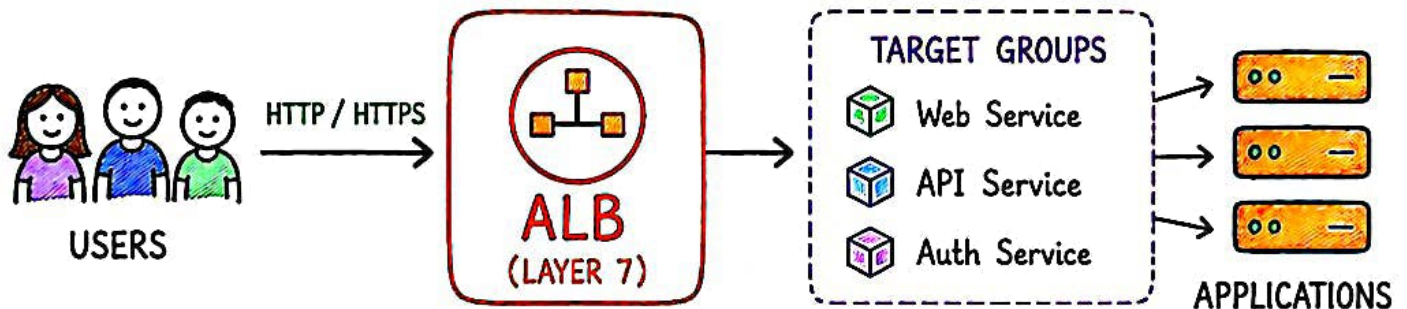
AWS LOAD BALANCING EXPLAINED
PRODUCTION + INTERVIEW READY



② APPLICATION LOAD BALANCER



ALB works at **LAYER 7** - APPLICATION LAYER
It understands HTTP/HTTPS and makes smart routing decisions.



KEY CAPABILITIES OF ALB



HTTP & HTTPS support
Handles web traffic natively.



Header & query-based routing
Route using headers, cookies or query parameters.



Host-based routing
Route traffic based on domain name.



WebSocket support
Ideal for real-time applications.



Path-based routing
Route traffic based on URL path.



Great for microservices
Distribute traffic across multiple services with ease.

EXAMPLE: HOST & PATH BASED ROUTING

shop.example.com

Shop Service

api.example.com

API Service

example.com/images/*

Image Service



ALB is Best For:

- ✓ Web applications
- ✓ Microservices architectures
- ✓ Content-based routing
- ✓ Multiple domains & paths

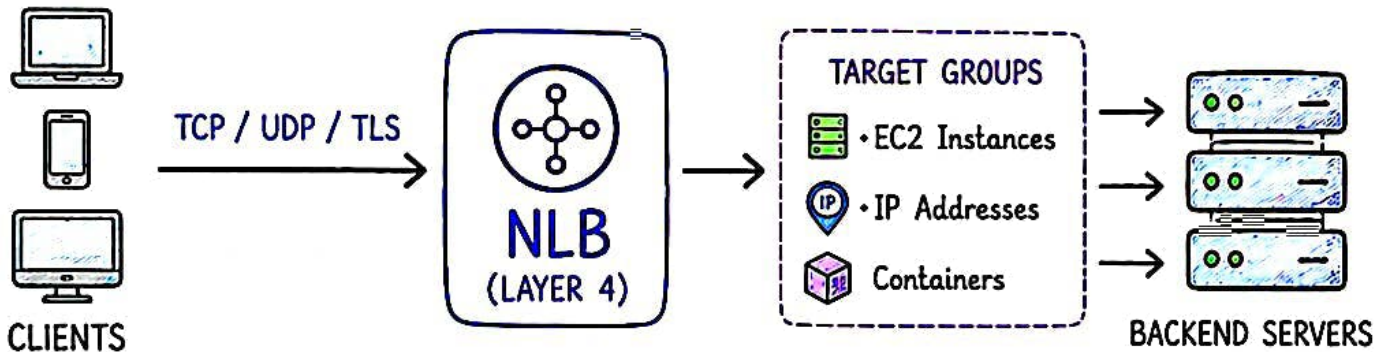


ALB UNDERSTANDS REQUEST CONTENT
AND **ROUTES TRAFFIC INTELLIGENTLY!**

3 NETWORK LOAD BALANCER



NLB works at **LAYER 4 - TRANSPORT LAYER**
It routes traffic based on IP addresses and ports.



KEY CAPABILITIES OF NLB



Ultra high performance
Handles millions of requests per second.



Supports TCP, UDP & TLS
Ideal for non-HTTP protocols, databases, VPN, gaming, etc.



Static IP support
Provides fixed IP addresses (Elastic IP).



High availability
Spread across multiple AZs for fault tolerance.



Preserves source IP
Backend servers see the original client IP.



Low latency
Designed for extreme performance and high concurrency.

WHEN TO USE NLB?



Databases
(RDS, MongoDB, MySQL)



VPN & Hybrid Apps
(TCP / UDP)



Gaming
Low latency, high performance



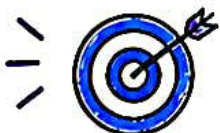
IoT Devices
Massive scale connections



Microservices
Communicating over TCP



High Performance
Apps needing extreme throughput



NLB DOESN'T LOOK AT THE CONTENT, IT MOVES THE CONNECTION FAST!

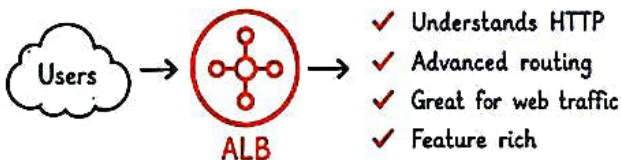
4 ALB vs NLB



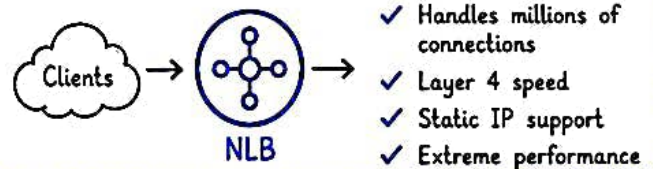
QUICK COMPARISON

FEATURE		ALB (LAYER 7)	NLB (LAYER 4)
	OSI LAYER	Layer 7 (Application Layer)	Layer 4 (Transport Layer)
	PROTOCOLS	HTTP, HTTPS	TCP, UDP, TLS
	ROUTING TYPE	Host-based, Path-based, Header & Query-based Routing	IP address & Port-based Routing
	CONTENT AWARE	✓ Yes (Understands HTTP Content)	✗ No (Does not look at Content)
	SSL/TLS TERMINATION	✓ Yes	✗ No* (TLS Passthrough Only)
	TARGET TYPE	IP, Instance, Lambda	IP, Instance
	STATIC IP	✗ No	✓ Yes (Supports Elastic IP)
	PERFORMANCE	High	Very High
	SCALABILITY	High	Very High
	COST	Pay per LCU (Capacity Units)	Pay per NLCU (Capacity Units)
	BEST FOR	Web apps, Microservices, Monoliths, HTTP routing	High performance, TCP/UDP apps, Real-time, Gaming, Databases, VPN, IoT

ALB → SMART APPLICATION ROUTING



NLB → HIGH PERFORMANCE NETWORK ROUTING



CHOOSE BASED ON YOUR PROTOCOL,
ROUTING NEEDS & PERFORMANCE REQUIREMENTS!

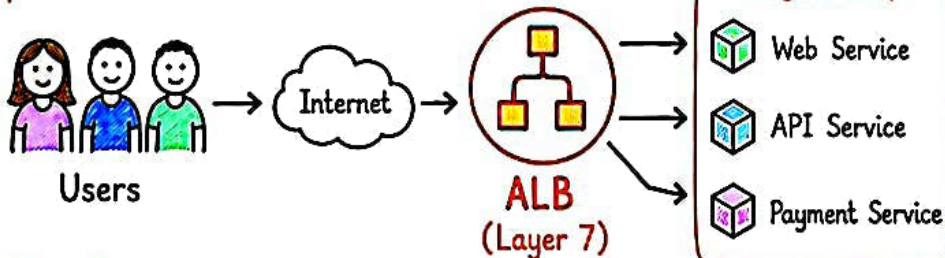
5 REAL PRODUCTION SCENARIO



WHICH ONE WOULD YOU USE?

A) USE ALB (LAYER 7) - WEB APPLICATION

Use when you need HTTP/HTTPS routing, host/path based routing



Example:

shop.example.com → Web Service
api.example.com → API Service
shop.example.com/pay → Payment Service



Best For:

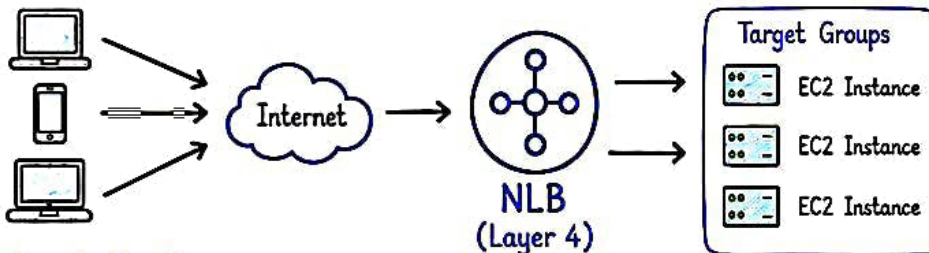
Web applications, Microservices, Path/Host based routing, Content based routing

Why ALB?

- ✓ HTTP/HTTPS
- ✓ Host based routing (shop.example.com)
- ✓ Path based routing (/api, /images)
- ✓ Advanced routing rules
- ✓ WebSocket support
- ✓ Ideal for Microservices

B) USE NLB (LAYER 4) - HIGH PERFORMANCE TCP APPLICATION

Use when you need TCP/UDP/TLS, static IPs, extreme performance



Example Use Cases:



Databases (RDS, MongoDB, MySQL)



Gaming Servers



VPN Connections



IoT / MQTT Applications



Real-time Streaming

Why NLB?

- ✓ TCP, UDP, TLS
- ✓ Very high performance
- ✓ Static IP (Elastic IP)
- ✓ Preserves source IP
- ✓ Handles millions of connections
- ✓ Ideal for extreme network workloads

KEY TAKEAWAY



Choose **ALB** for intelligent application routing.
Choose **NLB** for extreme performance and network workloads.



Remember:

Protocol + Routing Needs + Performance + IP requirements
decide your choice, not just the traffic volume!

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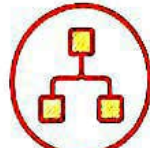
ALB OR NLB ?

INTERVIEW CHEAT SHEET



CHOOSE ALB WHEN YOU NEED:

- ✓ HTTP/HTTPS traffic
- ✓ Host based routing
- ✓ Path based routing
- ✓ Header & query based routing
- ✓ WebSocket support
- ✓ Advanced routing rules
- ✓ Microservices & Web applications (Layer 7)
- ✓ Application layer intelligence

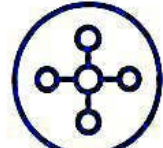


ALB

(Layer 7)

CHOOSE NLB WHEN YOU NEED:

- ✓ TCP, UDP, TLS traffic
- ✓ Very high performance
- ✓ Static IP (Elastic IP)
- ✓ Preserve source IP
- ✓ Extreme scale & low latency
- ✓ Million of connections
- ✓ Non-HTTP workloads
- ✓ Layer 4 network routing



NLB

(Layer 4)

⇒ QUICK DECISION GUIDE ⇐

IF YOUR APPLICATION...	THEN CHOOSE...	BECAUSE...
Needs HTTP routing, host/path rules	ALB	It understands HTTP and routes intelligently
Needs extreme performance (TCP/UDP)	NLB	It is built for speed at Layer 4
Needs static IP addresses	NLB	It supports Elastic IP addresses
Is a microservices / web application	ALB	It provides advanced application level routing
Handles millions of concurrent connections	NLB	It scales better for network level traffic

⇒ INTERVIEW QUESTION ⇐



Your application receives 100,000 requests per second.
Would you choose ALB or NLB?



BEST ANSWER: Traffic volume alone isn't enough to decide.
First understand your protocol (HTTP/TCP/UDP), routing requirements, performance needs, and IP requirements. Then choose ALB or NLB.



REMEMBER:

Few VPCs → VPC Peering

Many VPCs → Transit Gateway



SAVE THIS POST FOR YOUR AWS INTERVIEW PREPARATION!



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